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Part 1

Project Overview

The Problem included reviving an old 2009 project (Financial Calculator). This needed extensive modernization and compilation to ensure it runs correctly.

Bug fixes and modernization: Fix serious compilation and linking issues brought on by an antiquated build environment and inadequate dependency management. This involved correcting GoogleTest library include paths and updating the C++ standard. It also included making changes in the code which included giving the codes **Input Validation** , Fixing **potential integer overflow** in tenure calculations by using **long double** and safer math. It also included removing hardcoded values by **reading configuration from a file (config.txt)**.

Unit Testing Implementation: To validate the fundamental business logic, particularly the computation of loan payments and the validation of input parameters, integrate the GoogleTest framework and create at least three reliable unit tests.

Technical Documentation: Use Doxygen to provide thorough technical documentation that describes the operation and structure of the updated codebases.

The successful completion of this project transitioned the codebase from an unusable legacy artifact to a stable, quality-assured, and well-documented application. The successful execution of all unit tests and the generation of complete documentation confirm the project's success.

Part 2

Bug Fixes

Fix 1: Enforcing C++17 Standard

Before:

```
DELL@DESKTOP-79BCGFB MINGW64 ~/Desktop/ISEproject/legacy-calc-2009/src (dev)
$ g++ -std=c++11 \
-I$PROJECT_ROOT/googletest/googletest/include \
Loan.cpp Utils.cpp test/test_emi.cpp \
-o run_tests \
-L$PROJECT_ROOT/googletest/build/lib \
-lgtest_main -lgtest -lgtest -pthread
In file included from C:/Users/DELL/Desktop/ISEproject/legacy-calc-2009/googletest/googletest/include/gtest/gtest-message.h:57,
                 from C:/Users/DELL/Desktop/ISEproject/legacy-calc-2009/googletest/googletest/include/gtest/gtest-assertion-result.h:46,
                 from C:/Users/DELL/Desktop/ISEproject/legacy-calc-2009/googletest/googletest/include/gtest/gtest.h:63,
                 from test/test_emi.cpp:11:
C:/Users/DELL/Desktop/ISEproject/legacy-calc-2009/googletest/include/gtest/internal/gtest-port.h:260:2: error: #error C++ versions less than C++17 are not supported.
 260 | #error C++ versions less than C++17 are not supported.
```

After:

```
DELL@DESKTOP-79BCGFB MINGW64 ~/Desktop/ISEproject/legacy-calc-2009/src (dev)
$ g++ -std=c++17 \
-I$PROJECT_ROOT/googletest/googletest/include \
Loan.cpp Utils.cpp test/test_emi.cpp \
-o run_tests \
-L$PROJECT_ROOT/googletest/build/lib \
-lgtest_main -lgtest -lgtest -pthread
```

Fix 2: Correcting GoogleTest Header Include Path Before:

```
DELL@DESKTOP-79BCGFB MINGW64 ~/Desktop/ISEproject/legacy-calc-2009/src (dev)
$ g++ -std=c++11 \
-isystem $GTEST_PATH/include \
Loan.cpp Utils.cpp test/test_emi.cpp \
-o run_tests \
-L$GTEST_PATH/build/lib \
-lgtest_main -lgtest -lgtest -pthread
test/test_emi.cpp:1:10: fatal error: gtest/gtest.h: No such file or directory
 1 | #include "gtest/gtest.h"
   |           ^~~~~~
compilation terminated.
```

After:

```
DELL@DESKTOP-79BCGFB MINGW64 ~/Desktop/ISEproject/legacy-calc-2009/src (dev)
$ g++ -std=c++17 \
-I$PROJECT_ROOT/googletest/googletest/include \
Loan.cpp Utils.cpp test/test_emi.cpp \
-o run_tests \
-L$PROJECT_ROOT/googletest/build/lib \
-lgtest_main -lgtest -lgtest -pthread

DELL@DESKTOP-79BCGFB MINGW64 ~/Desktop/ISEproject/legacy-calc-2009/src (dev)
```

Fix 3: Adding Robust Input Validation

```

bool validateLoanParameters(long double principal, long double annual_rate, int tenure_months) {
    bool isValid = true;

    if (principal <= 0.0L) {
        std::cerr << "ERROR: Loan principal amount must be positive." << std::endl;
        isValid = false;
    }

    // Annual rate can be 0% but not negative
    if (annual_rate < 0.0L) {
        std::cerr << "ERROR: Annual interest rate cannot be negative." << std::endl;
        isValid = false;
    }

    if (tenure_months <= 0) {
        std::cerr << "ERROR: Loan tenure must be a positive number of months." << std::endl;
        isValid = false;
    }

    return isValid;
}

```

Fix 3: Preventing Integer overflow with Long double

```

long double LoanCalculator::calculatePayment()
{
    if(!amountSet_ || !interestSet_ || !periodTotalSet_)
    {
        throw invalid_argument("Must set loan amount, interest, and total period for this calculation" );
    }

    long double totalAmount = amount_ - initialPayment_;
    totalAmount = totalAmount + openingFee_ + (totalAmount * (openingPercent_/100.0L)); // Added L to 100.0

    if (interestPeriodic_ == 0.0L) {
        if (periodTotal_ == 0) {
            throw invalid_argument("Loan period cannot be zero." );
        }
        return totalAmount / periodTotal_;
    }

    // Use powl() for high precision
    long double numerator = interestPeriodic_ * totalAmount;
    long double denominator = 1.0L - powl((1.0L + interestPeriodic_), (-1 * periodTotal_));

    return numerator / denominator;
}

```

Fix 3: Reading Configuration from File

```

int getOutputPrecision() {
    std::ifstream configFile("config.txt");
    std::string line;
    const std::string key = "OUTPUT_PRECISION=";
    int precision = 2; // Default value if reading fails

    if (configFile.is_open()) {
        while (std::getline(configFile, line)) {
            // Check if line starts with the key
            if (line.rfind(key, 0) == 0) {
                std::string valueStr = line.substr(key.length());
                try {
                    // Safely convert string to integer
                    precision = std::stoi(valueStr);
                    // Basic sanity check for precision
                    if (precision < 0 || precision > 15) {
                        precision = 2;
                    }
                } catch (const std::exception& e) {
                    std::cerr << "Warning: Invalid precision value in config.txt. Using default (2).\n";
                    configFile.close();
                    return 2;
                }
            }
        }
        configFile.close();
    } else {
        // This warning is helpful during debugging
        std::cerr << "Warning: config.txt not found. Using default output precision (2).\n";
    }

    return precision;
}

```

CodeBase Refactoring:

Refactoring 1: Decomposing main.cpp into Modules

This PC > Desktop > ISEproject > legacy-calc-2009 > src				
	Name	Date modified	Type	Size
ss	Loan	11/25/2025 1:51 AM	C++ Source File	
ls	Loan	11/25/2025 7:03 PM	C Header File	
ts	main	11/25/2025 7:04 PM	C++ Source File	
	Utils	11/25/2025 6:57 PM	C++ Source File	
	Utils	11/25/2025 1:41 AM	C Header File	

Refactoring 2: Improving Output Formatting and Consistency

```

// Output result
std::cout << "\n--- Loan Calculation Summary ---" << std::endl;
std::cout << std::fixed << std::setprecision(precision);
std::cout << "Loan Amount:    $" << calc.getAmount() << std::endl;
std::cout << "Annual Rate:    " << calc.getInterest() << "%" << std::endl;
std::cout << "Total Periods:  " << calc.getPeriodTotal() << " months" << std::endl;
std::cout << "Monthly Payment: $" << emi << std::endl;
std::cout << "-----" << std::endl;

```

Step 3

Screen shots of test output:

```

DELL@DESKTOP-79BCGFB MINGW64 ~/Desktop/ISEproject/legacy-calc-2009 (dev)
$ ./googletest/build/run_tests
Running main() from C:/Users/DELL/Desktop/ISEproject/legacy-calc-2009/googletest/googletest/src/gtest_main.cc
[-----] Running 3 tests from 2 test suites.
[-----] Global test environment set-up.
[-----] 2 tests from LoanCalculatorTest
[ RUN      ] LoanCalculatorTest.CalculateStandardPayment
[ OK       ] LoanCalculatorTest.CalculateStandardPayment (0 ms)
[ RUN      ] LoanCalculatorTest.LargeTenurePayment
[ OK       ] LoanCalculatorTest.LargeTenurePayment (0 ms)
[-----] 2 tests from LoanCalculatorTest (13 ms total)

[-----] 1 test from UtilsTest
[ RUN      ] UtilsTest.ValidateLoanParametersInvalid
ERROR: Loan principal amount must be positive.
ERROR: Annual interest rate cannot be negative.
ERROR: Loan tenure must be a positive number of months.
ERROR: Loan principal amount must be positive.
ERROR: Loan tenure must be a positive number of months.
ERROR: Annual interest rate cannot be negative.
[ OK       ] UtilsTest.ValidateLoanParametersInvalid (1 ms)
[-----] 1 test from UtilsTest (7 ms total)

[-----] Global test environment tear-down
[=====] 3 tests from 2 test suites ran. (41 ms total)
[ PASSED ] 3 tests.

```

Step 4

Git commit history:

```

DELL@DESKTOP-79BCGFB MINGW64 ~/Desktop/ISEproject/legacy-calc-2009 (dev)
$ git log --oneline --graph
* 068e435 (HEAD -> dev) Docs: Added final project report, all requirements met.
* 8d81cff Setup: Complete Assignment 2 setup. Added .gitignore, updated README, and created dev branch.
* d54b9f3 (origin/dev) Setup project structure: src, test, docs, .gitignore, config.txt
* 347e865 (origin/master, origin/HEAD, master) Added compilation information
* dff0427 Nicer formatting on Readme and added better app help text
* 71f0456 Added readme and additional app help text
* c3e0801 Added a Qt Gui and scons compilation. The makefile was generated by Qt.
* be10569 Changes to use libCmdLineParser
* 12ee333 Removed compileCommands.txt
* 4d09e8b Made methods inline and removed some compiler errors
* d99e446 First version

```